

It has the following structure:

Date	DAP	SAP
2014-06-12	90	130
2014-07-04	92	130
2014-07-27	85	125
...	...	...

The final result is shown in Figure 1.

2D plot examples

Using the above mentioned meteorological data, an area plot can be easily built. In the axis section, the following options are used:

```
p1/.style={draw=DodgerBlue,draw opacity=0.6,line
width=3pt,fill=DodgerBlue,fill opacity=0.6,mark=none},
p2/.style={draw=Salmon,draw opacity=0.6,line
width=3pt,fill=Salmon,fill opacity=0.6,mark=none}
```

And then each data series is plotted, as follows:

```
\addplot [p1] table[x=M,y=Milan]{\datarain}\closedcycle;
\addplot [p2] table[x=M,y=Palermo]{\datarain}\closedcycle;
```

The option *closedcycle* defines a closed surface for each series. This is the main difference between an area plot and a bar plot. The same table could also be plotted as a bar plot. In the preamble, the following lines are added to have the height of each bar written on the plot considering the first decimal figure, even if it's a zero.

```
\pgfkeys{
/pgf/number format/precision=1,
/pgf/number format/fixed zerofill=true}
```

The *axis* section contains, among the other options, the following code. The first three lines are to write on the plot the height of each bar and the last two to select the colour of each bar type.

```
nodes near coords,
nodes near coords align={vertical},
every node near coord/.append style={font=\tiny},
...
p1/.style={draw=DodgerBlue,fill=DodgerBlue},
p2/.style={draw=Salmon,fill=Salmon}
```

Last of all, each series is plotted, as shown below:

```
\addplot [p1] table[x=M,y=Milan]{\datarain};
\addplot [p2] table[x=M,y=Palermo]{\datarain};
```

On the same plot, it's possible to add two (or more) data series with different plot types. For example, I can plot the

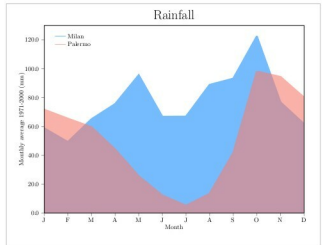


Figure 2: An area plot

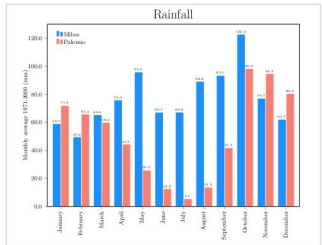


Figure 3: A bar plot

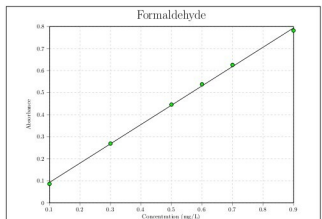


Figure 4: A scatter plot

above mentioned *coordinates* table as points only, as follows:

```
\addplot[only marks,mark=*,mark size=3pt,mark
options={fill=green}]
coordinates {...};
```